

MH2100 Calculus III

Midterm I (Solutions)

AY 2018/2019, Semester 1

Quantitative Research Society @NTU

September 2018

Overview

- **Coverage:** parametric curves and vector calculus; partial derivatives, limits, continuity and differentiability; critical points and the second-derivative test.
- **Format:** three questions, 100 marks, 90 minutes (official paper: September 2018).
- **Historical permitted material:** the September 2018 paper permitted a two-sided A4 piece of paper with handwritten notes. Consult current course materials for present assessment rules.

Instructions (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. A two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(50 Marks)**

Circle (only) the correct answer for each question. Each question is worth 5 marks.

- (i) Find parametric equations to represent the line segment from $(-3, 4)$ to $(12, -8)$.

- (a) $x = -3 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1$
- (b) $x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$
- (c) $x = 8 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$
- (d) $x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1$
- (e) $x = -3 - 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 2$

- (ii) Find the point(s) on the following curve where the tangent is horizontal.

$$x = t^3 - 3t + 2, \quad y = t^3 - 3t^2 + 2$$

- (a) $(0, 0)$
- (b) $(0, 0)$ and $(2, -2)$
- (c) $(2, 2)$ and $(-2, -2)$
- (d) $(2, 2)$ and $(4, -2)$
- (e) None of the above

- (iii) Find parametric equations for the tangent line at the point when $t = 1$ to the curve with parametric equations

$$x = 3t, \quad y = 5t^2, \quad z = 4t^3.$$

- (a) $x = 1 + 3t, \quad y = 1 + 10t, \quad z = 1 + 12t$
- (b) $x = 3 + 3t, \quad y = 5 + 10t, \quad z = 4 + 12t$
- (c) $x = 1 + 3t, \quad y = 1 + 5t, \quad z = 1 + 4t$
- (d) $x = 3 + 3t, \quad y = 5 + 5t, \quad z = 4 + 4t$
- (e) None of the above

- (iv) Find the unit tangent vector for the curve given by

$$\mathbf{r}(t) = \left\langle \frac{1}{5}t^5, \frac{1}{3}t^3, t \right\rangle.$$

- (a) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^{10} + t^4}}$
- (b) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{4t^8 + 4t^4 + 1}}$
- (c) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4 + 1}}$
- (d) $\frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4}}$
- (e) None of the above

- (v) Find the directional derivative of the function $f(x, y, z) = xyz$ in the direction of the vector $\langle 5, -3, 2 \rangle$.

- (a) $\frac{1}{\sqrt{38}}(5yz - 3xz + 2xy)$
- (b) $\frac{1}{\sqrt{38}}(5xy - 3xz + 2yz)$
- (c) $5yz - 3xz + 2xy$
- (d) $5xy - 3xz + 2yz$
- (e) None of the above
- (vi) Find the gradient vector of the function $f(x, y, z) = z^7 e^{2x\sqrt{y}}$.
- (a) $e^{x^2\sqrt{y}} \left\langle 7xz\sqrt{y}, \frac{x^2 z}{2\sqrt{y}}, 2 \right\rangle$
- (b) $ze^{2x\sqrt{y}} \left\langle z\sqrt{y}, \frac{xz}{2\sqrt{y}}, 7z^5 \right\rangle$
- (c) $e^{x^2\sqrt{y}} \left\langle 2xz\sqrt{y}, \frac{x^2 z}{7\sqrt{y}}, 2 \right\rangle$
- (d) $ze^{2x\sqrt{y}} \left\langle 2z^6\sqrt{y}, \frac{xz^6}{\sqrt{y}}, 7z^5 \right\rangle$
- (vii) Use the Chain Rule to find $\frac{dw}{dt}$ for
- $$w = 3x^6 y^3 z, \quad x = 6t, \quad y = \cos 2t, \quad z = t \sin t.$$
- (a) $3x^5 y^2 (36yz - 6xz \sin 2t + xy(\sin t + t \cos t))$
- (b) $3x^6 y^3 (36yz - 6xz \sin 2t + xy(\sin t + t \cos t))$
- (c) $648x^5 y^2 \sin 2t (\sin t + t \cos t)$
- (d) $648x^6 y^3 \sin 2t (\sin t + t \cos t)$
- (e) None of the above
- (viii) Find an equation for the tangent plane to the surface given by the equation $xy + yz + xz = 38$ at the point $(4, 2, 5)$.
- (a) $7x + 9y + 6z = 70$
- (b) $7x + 9y + 6z = 74$
- (c) $7x + 9y + 6z = 76$
- (d) $7x + 9y + 6z = 78$
- (e) None of the above
- (ix) Find an equation for the tangent plane to the surface $z = 4x^3 y^2 + 2y$ at the point $(1, -2, 12)$.
- (a) $48x + 14y - z = 60$
- (b) $48x - 14y - z = 64$
- (c) $48x - 14y - z = 60$
- (d) $48x + 14y - z = 64$

- (e) None of the above
- (x) Find and classify the critical points of the function $f(x, y) = e^{-2x} \sin 4y$ for $x \geq 0$ and $0 \leq y \leq \pi/2$.
- (a) Local maximum $f(0, \pi/8) = 1$
- (b) Saddle point $f(0, 0) = 0$
- (c) Local minimum $f(0, \pi/4) = 0$
- (d) None of the above

Solution

Method 1: Direct computation

- (i) A segment from $\mathbf{p} = (-3, 4)$ to $\mathbf{q} = (12, -8)$ is $\mathbf{p} + t(\mathbf{q} - \mathbf{p})$, $0 \leq t \leq 1$. Since $\mathbf{q} - \mathbf{p} = (15, -12)$,

$$(d) \quad x = -3 + 15t, \quad y = 4 - 12t, \quad 0 \leq t \leq 1.$$

- (ii) $y' = 3t(t - 2)$ and $x' = 3(t^2 - 1)$. Horizontal tangents occur at $t = 0, 2$, where $x' \neq 0$. The points are $(2, 2)$ and $(4, -2)$:

$$(d) \quad (2, 2) \text{ and } (4, -2).$$

- (iii) $\mathbf{r}(1) = (3, 5, 4)$ and $\mathbf{r}'(1) = (3, 10, 12)$. Thus

$$(b) \quad x = 3 + 3t, \quad y = 5 + 10t, \quad z = 4 + 12t.$$

- (iv) $\mathbf{r}'(t) = \langle t^4, t^2, 1 \rangle$ and $\|\mathbf{r}'(t)\| = \sqrt{t^8 + t^4 + 1}$, so

$$(c) \quad \mathbf{T}(t) = \frac{\langle t^4, t^2, 1 \rangle}{\sqrt{t^8 + t^4 + 1}}.$$

- (v) $\nabla f = \langle yz, xz, xy \rangle$ and the unit direction is $\langle 5, -3, 2 \rangle / \sqrt{38}$. Hence

$$(a) \quad D_{\mathbf{u}}f = \frac{5yz - 3xz + 2xy}{\sqrt{38}}.$$

- (vi) On the natural interior of the real domain, where $y > 0$, differentiating each variable gives

$$\nabla f = e^{2x\sqrt{y}} \left\langle 2z^7\sqrt{y}, \frac{xz^7}{\sqrt{y}}, 7z^6 \right\rangle = z e^{2x\sqrt{y}} \left\langle 2z^6\sqrt{y}, \frac{xz^6}{\sqrt{y}}, 7z^5 \right\rangle.$$

$$(d).$$

- (vii) The chain rule gives

$$\begin{aligned} \frac{dw}{dt} &= w_x(6) + w_y(-2 \sin 2t) + w_z(\sin t + t \cos t) \\ &= 3x^5y^2(36yz - 6xz \sin 2t + xy(\sin t + t \cos t)). \end{aligned}$$

$$(a).$$

(viii) For $F = xy + yz + xz$, $\nabla F(4, 2, 5) = \langle 7, 9, 6 \rangle$. Therefore

$$7(x - 4) + 9(y - 2) + 6(z - 5) = 0, \quad \boxed{\text{(c)} \quad 7x + 9y + 6z = 76.}$$

(ix) $f_x = 12x^2y^2$ and $f_y = 8x^3y + 2$, so at $(1, -2)$ they are 48 and -14 . The tangent plane is

$$z - 12 = 48(x - 1) - 14(y + 2), \quad \boxed{\text{(b)} \quad 48x - 14y - z = 64.}$$

(x) **Literal versus intended reading.** Under the standard definition, a critical point of a differentiable function of two variables must be an interior point of its domain where $\nabla f = 0$. Here

$$f_x = -2e^{-2x} \sin 4y, \quad f_y = 4e^{-2x} \cos 4y,$$

and these cannot vanish simultaneously. Hence the literal critical-point reading gives no critical points and would select **(d)**.

The apparent intended reading is to classify relative extrema on the stated constrained domain, including its boundary. At $(0, \pi/8)$, $f = 1$; nearby in the domain, $e^{-2x} \leq 1$ and $\sin 4y \leq 1$, so it is a relative local maximum. Thus the intended multiple-choice answer is

$$\boxed{\text{intended: (a) Local maximum } f(0, \pi/8) = 1; \quad \text{literal: (d).}}$$

Unofficial QRS marking scheme (50 marks)

- (i) **(5 marks)** 2: correct displacement; 2: correct parameter interval; 1: selects (d).
- (ii) **(5 marks)** 2: $y' = 0$; 2: checks $x' \neq 0$ and finds both points; 1: selects (d).
- (iii) **(5 marks)** 2: point; 2: tangent vector; 1: selects (b).
- (iv) **(5 marks)** 2: derivative; 2: norm; 1: selects (c).
- (v) **(5 marks)** 2: gradient; 2: normalized direction; 1: selects (a).
- (vi) **(5 marks)** 4: all gradient components; 1: selects (d).
- (vii) **(5 marks)** 4: correct chain-rule terms and simplification; 1: selects (a).
- (viii) **(5 marks)** 2: normal; 2: plane through the point; 1: selects (c).
- (ix) **(5 marks)** 2: partial derivatives; 2: plane; 1: selects (b).
- (x) **(5 marks)** 2: identifies the boundary-definition issue; 2: constrained local-maximum argument; 1: states intended (a) and literal (d).

Question 2**(25 Marks)**

- (i) Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a) $\lim_{(x,y) \rightarrow (6,6)} \frac{5e^{x-y}}{\cos(\pi(x-y))}$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^{21}y^{21}}{x^{42} + y^{84}}$

- (ii) Define

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{x^2 + y^2}\right); & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct limit and derivative definitions**

- (i) (a) At $(6, 6)$ the denominator is $\cos 0 = 1 \neq 0$. The quotient is continuous there, hence

$$\lim_{(x,y) \rightarrow (6,6)} \frac{5e^{x-y}}{\cos(\pi(x-y))} = 5.$$

- (b) Along $y = 0$, the quotient is 0. Along $y = x$ with $x \neq 0$ it is

$$\frac{x^{42}}{x^{42} + x^{84}} = \frac{1}{1 + x^{42}} \rightarrow 1.$$

The two path limits differ, so

the limit does not exist.

- (ii) (a) Since $|\sin u| \leq 1$,

$$|f(x, y)| \leq x^2 + y^2 \rightarrow 0 = f(0, 0).$$

Thus

f is continuous at $(0, 0)$.

- (b) By the partial-derivative definitions,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{h^2 \sin(1/h^2)}{h} = \lim_{h \rightarrow 0} h \sin(1/h^2) = 0,$$

and identically $f_y(0, 0) = 0$. Hence

$$f_x(0, 0) = f_y(0, 0) = 0.$$

- (c) The only candidate differential is $L(h, k) = 0$. With $r = \sqrt{h^2 + k^2}$,

$$\frac{|f(h, k) - f(0, 0) - L(h, k)|}{r} = r \left| \sin\left(\frac{1}{r^2}\right) \right| \leq r \rightarrow 0.$$

Therefore

f is differentiable at $(0, 0)$, $Df(0, 0) = 0$.

Unofficial QRS marking scheme (25 marks)**(i) (10 marks)**

- (a) **(5 marks)** 2: recognizes continuity of the quotient; 2: evaluates numerator and denominator; 1: final value 5.
- (b) **(5 marks)** 2: path $y = 0$; 2: path $y = x$; 1: concludes nonexistence.

(ii) (15 marks)

- (a) **(5 marks)** 3: valid squeeze bound; 2: continuity conclusion.
- (b) **(5 marks)** 2 each: correct difference quotients; 1: both values equal zero.
- (c) **(5 marks)** 2: identifies $L = 0$; 2: normalizes the remainder; 1: differentiability conclusion.

Question 3**(25 Marks)**

Let

$$f(x, y) = y^3 - x^2 - 7y^2 + 2xy + 9y.$$

Find and classify all critical points of f . (For each critical point, determine whether or not it is a local max/min or a saddle point.)

Solution**Method 1: Direct derivative and second-derivative test**

$$f_x = -2x + 2y, \quad f_y = 3y^2 - 14y + 2x + 9.$$

The equation $f_x = 0$ gives $x = y$. Substitution into $f_y = 0$ gives

$$3y^2 - 12y + 9 = 3(y - 1)(y - 3) = 0.$$

Thus the critical points are $(1, 1)$ and $(3, 3)$. The Hessian entries are

$$f_{xx} = -2, \quad f_{xy} = 2, \quad f_{yy} = 6y - 14,$$

so the second-derivative discriminant is

$$D = f_{xx}f_{yy} - f_{xy}^2 = (-2)(6y - 14) - 4 = 24 - 12y.$$

At $(1, 1)$, $D = 12 > 0$ and $f_{xx} < 0$, so it is a local maximum (with $f(1, 1) = 4$). At $(3, 3)$, $D = -12 < 0$, so it is a saddle point (with $f(3, 3) = 0$). Therefore

$(1, 1)$ is a local maximum; $(3, 3)$ is a saddle point.
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Method 2: Completing the square and local signs

Completing the square in x gives

$$f(x, y) = y(y - 3)^2 - (x - y)^2.$$

Near $(1, 1)$, put $u = x - y$ and $v = y - 1$. Then

$$f(x, y) = 4 - u^2 - 3v^2 + v^3.$$

For $|v| < 1$, the last two terms satisfy $-3v^2 + v^3 = -v^2(3 - v) \leq -2v^2$, so $(1, 1)$ is a strict local maximum. Near $(3, 3)$, put $u = x - y$ and $v = y - 3$. Then

$$f(x, y) = 3v^2 + v^3 - u^2.$$

Along $u = 0$ this is positive for small nonzero v , while along $v = 0$ it is negative for nonzero u . Hence $(3, 3)$ is a saddle point.

Unofficial QRS marking scheme (25 marks)

The breakdown follows Method 1; an equivalent rigorous structural proof, such as Method 2, receives the corresponding full credit.

- **8 marks:** computes f_x, f_y and correctly solves the simultaneous equations.
- **5 marks:** identifies both critical points $(1, 1)$ and $(3, 3)$.
- **6 marks:** computes the Hessian entries and discriminant correctly.
- **3 marks:** applies the test to classify $(1, 1)$ as a local maximum.
- **3 marks:** applies the test to classify $(3, 3)$ as a saddle point.